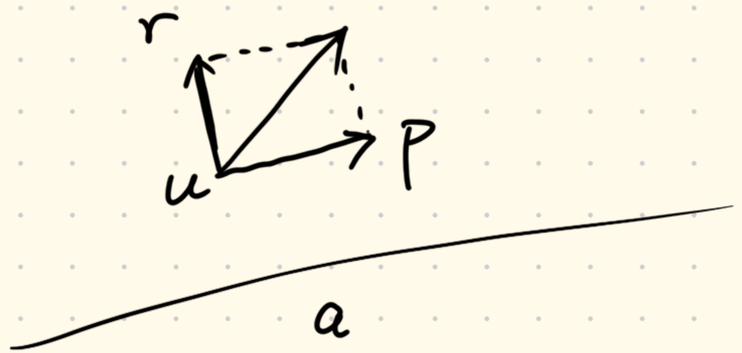
$$(a|b) = \sum a_i b_i = a^T b = b^T a.$$

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$



orthogonal projection

$a \neq 0$, want to write $u = p + r$, $p \parallel a$, $r \perp a$.

$$\rightsquigarrow 0 = (u - ta|a) = (u|a) - t(a|a) \Rightarrow t = \frac{(u|a)}{(a|a)}.$$

$$\text{write } p = Pu. \quad p = \frac{(u|a)}{(a|a)} a = \frac{(a^T u)a}{a^T a} = \frac{a(a^T u)}{a^T a}$$

$$= \frac{a a^T}{a^T a} u \rightsquigarrow P = \frac{a a^T}{a^T a} \text{ orthogonal matrix.}$$

$$P^T = \frac{(aa^T)^T}{a^Ta} = \frac{aa^T}{a^Ta} = P.$$

P is idempotent, $P^2 = \frac{1}{(a^Ta)^2}(aa^T)(aa^T)$

$$= \frac{1}{(a^Ta)^2}\, a(a^Ta)a^T = \frac{aa^T}{a^Ta} = P.$$

e.g. $u = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $P = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$P = \begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{pmatrix}.$$

Then $u = p + r$.

$$\|u\|^2 = \|p+r\|^2 = \|p\|^2 + \|r\|^2 = \frac{(u|a)^2}{(a|a)^2}\|a\|^2 + \|r\|^2$$

$$= \frac{(u|a)^2}{\|a\|^2} + \|r\|^2 \geq \frac{(u|a)^2}{\|a\|^2} \quad \Rightarrow \quad (u|a)^2 \leq \|u\|^2 \cdot \|a\|^2.$$

$r = 0 \Leftrightarrow u \parallel a.$

Cauchy - Schwarz inequality

In $\mathbb{R}^n$. Euclidean inner product. $\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)$

equality holds when a.b linear dependent. $\left(\sum_{i=1}^{n} b_i^2\right)$.

$\{f \in C^0(\mathbb{R}^n, \mathbb{R})\}$, $\langle f \cdot g \rangle = \int_a^b fg$,

$$\left|\int_a^b fg\right|^2 \leq \left(\int_a^b f\right)\left(\int_a^b g\right).$$

matrix space $M_{m\times n}(\mathbb{R})$, Frobenius inner product

$\langle A, B\rangle_F = tr(A^T B)$. $\quad M_{m\times n}(\mathbb{R}) \cong \mathbb{R}^{mn}$. ✓

$tr(A^T B)^2 \le tr(A^T A)\cdot tr(B^T B)$

$$\left|\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ij}\right|^2 \le \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right)\cdot\left(\sum_{i=1}^{m}\sum_{j=1}^{n} b_{ij}^2\right).$$

(Least Squares Problem).

$x \in \mathbb{R}^n$, find a point p in $W = span(v_1, \cdots, v_m)$ s.t.

$\|x-p\|$ is minimized. $\quad$ solve $\min_{p\in W} \|x-y\|$.

$P = P_X$. $P = A(A^TA)^{-1}A^T$, where $A = (v_1 | \cdots | v_m)$

$\in M_{m\times n}(\mathbb{R})$, assume $v_1, \cdots, v_m$ are linearly independent.

Inner product Space, V is a vector space endowed with an inner product. $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$. $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

satisfying:

1. (conjugate symmetry) $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

2. (linearity) $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.

3. (Positive-definiteness) $\langle x, x \rangle \geq 0$, $\langle x, x \rangle = 0 \Leftrightarrow x = 0$.

Cross product. $u = (u_1, u_2, u_3)^T$, $v = (v_1, v_2, v_3)^T$ in $\mathbb{R}^3$

$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u|v)^2$$

$$\Rightarrow \|u \times v\| = \|u\| \|v\| \sin\theta$$

torque

力矩 $\tau = r \times F$. turning effect of a force

angular momentum

角动量 $L = r \times p$

magnetic part of the Lorentz force $F_{mag} = q\, v \times B$.

magnetic force on a current element $dF = I\, dl \times B$.

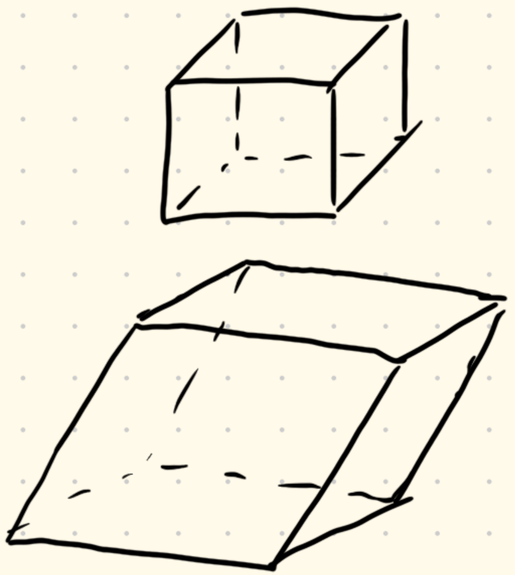
$$u = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \quad v = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \quad u \times v = \begin{pmatrix} 6 \\ -3 \\ 1 \end{pmatrix} \quad \|u \times v\| = \sqrt{46}.$$

Prop. $(\alpha u_1 + \beta u_2) \times v = \alpha (u_1, v) + \beta (u_2, v)$.

$u \times v = - v \times u$.

$u, v, w \in \mathbb{R}^3$. volumn $= (u \times v)^T w$

$= \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{pmatrix}$. orientation.



Determinant. In $\mathbb{R}^n$, an ordered list of n edges should determine a signed volumn.

Def. D is the unique function $D : (\mathbb{R}^n)^n \longrightarrow \mathbb{R}$, satisfying

multilinearity : $D(\cdots, \alpha u + \beta v, \cdots) = \alpha D(\cdots, u, \cdots) + \beta D(\cdots, v, \cdots)$.

alternation : $D(\cdots v_i \cdots v_j \cdots) = -D(\cdots v_j \cdots, v_i, \cdots)$.

normalization : $D(e_1, \cdots, e_n) = 1$.

$\det A := D(v_1, \cdots, v_n)$ column vectors.

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

$$D_n(A) = \sum_{j=1}^{n} (-1)^{1+j}\, a_{1j}\, D_{n-1}(A^{(1,j)}).$$

Prove $\det(AB) = \det A \cdot \det B$.

Define $F(v_1, \cdots, v_n) = \det(Av_1, \cdots, Av_n)$

is multilinear. alternating. By the uniqueness.

$F(v_1, \cdots, v_n) = \lambda \det(v_1, \cdots, v_n)$. $\lambda \in \mathbb{R}$.

Let $v_i = e_i$, $\lambda = \det(Ae_1, \cdots, Ae_n)$. $A = (u_1 | \cdots | u_n)$.

$= \det(u_1, \cdots, u_n) = \det A$.

$F(v_1, \cdots, v_n) = \det A \cdot \det(v_1, \cdots v_n) = \det(Av_1 \cdots Av_n)$

Let v_i be the columns of B. Then $\det(AB) = \det A \det B$.

3.2 8.
$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & 0 & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 & 0 \\ a_{51} & a_{52} & 0 & 0 & 0 \end{vmatrix} = 0.$$

3.3. 2(2)
$$\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$
$C_3 \leftarrow C_3 + (a+b+c)C_2 - (ab+bc+ca)C_1.$

(3)
$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (ab+bc+ca)\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$
$C_3 \leftarrow C_3 - (a+b+c)C_2 - abc\,C_1.$

3.3 (2.(1)

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ -1 & 0 & 3 & \cdots & n-1 & n \\ \vdots & -2 & 0 & \cdots & n-1 & n \\ \vdots & \vdots & -3 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 & n \\ -1 & -2 & -3 & \cdots & -(n-1) & 0 \end{vmatrix} = n! \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & 0 & 1 & \cdots & 1 \\ -1 & -1 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & 0 \end{vmatrix}$$

$C_j \longrightarrow C_j - C_{j-1}$

upper tragulen

$$\begin{vmatrix} 1 & 1 & & & \\ 0 & 1 & 1 & & \\ 0 & 0 & 1 & 1 & \\ & & & \ddots & \\ & & & & 1 \ 1 \\ & & & & 0 \ 1 \end{vmatrix} = 1.$$

3.3 1.

$$\begin{vmatrix} 1 & 6 & 5 & 3 \\ 2 & 5 & 8 & 1 \\ 3 & 4 & 5 & 1 \\ 4 & 5 & 8 & 2 \end{vmatrix}$$

$C_4 \leftarrow C_4 + 1000C_1 + 100C_2 + 10C_3$

$$\begin{vmatrix} 1 & 6 & 5 & 1653 \\ 2 & 5 & 8 & 2581 \\ 3 & 4 & 5 & 3451 \\ 4 & 5 & 8 & 4582 \end{vmatrix} = 29 \cdot ?$$